

Chemistry

CYL1010



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Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman, UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

Contents

- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Electrochemical Systems [5 Lectures]:** Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion



Applications of Cells

Secondary cells

Battery Type	Anode (oxidation)	Cathode (reduction)	Oxidized Form	Reduced Form	Electrolyte	Usages
Lead Acid (Pb-acid)	$\text{Pb} \rightarrow \text{PbSO}_4$	$\text{PbO}_2 \rightarrow \text{PbSO}_4$	Pb^{2+} (as PbSO_4)	Pb (metal)	H_2SO_4 (dilute sulfuric acid)	Cars, UPS, emergency lighting
Nickel-Cadmium (Ni-Cd)	$\text{Cd} \rightarrow \text{Cd(OH)}_2$	$\text{NiO(OH)} \rightarrow \text{Ni(OH)}_2$	Cd^{2+} (as Cd(OH)_2)	Ni(OH)_2	KOH (aq)	Power tools, emergency lighting
Nickel-Metal Hydride (Ni-MH)	$\text{MH} \rightarrow \text{M} + \text{H}^+$	$\text{NiO(OH)} \rightarrow \text{Ni(OH)}_2$	M (metal)	Ni(OH)_2	KOH (aq)	Hybrid cars, cameras
Lithium-ion (Li-ion)	$\text{LiC}_6 \rightarrow \text{C} + \text{Li}^+$	$\text{LiCoO}_2 \rightarrow \text{Li}_{1-x}\text{CoO}_2$	Li^+ (in electrolyte)	Li metal (in anode)	Organic solvent with Li salt	Phones, laptops, EVs
Lithium Polymer (Li-Po)	$\text{LiC}_6 \rightarrow \text{C} + \text{Li}^+$	$\text{LiMn}_2\text{O}_4 \rightarrow \text{Li}_{1-x}\text{Mn}_2\text{O}_4$	Li^+	Li^0	Polymer gel with Li salt	Drones, RC toys
Sodium-Nickel Chloride (Zebra)	$\text{Na} \rightarrow \text{Na}^+$	$\text{NiCl}_2 \rightarrow \text{Ni}$	Na^+	Ni	Molten NaAlCl_4	Grid storage, buses
Nickel-Zinc (Ni-Zn)	$\text{Zn} \rightarrow \text{Zn(OH)}_2$	$\text{NiO(OH)} \rightarrow \text{Ni(OH)}_2$	Zn^{2+}	Ni(OH)_2	KOH (aq)	Tools, cameras
Silver-Zinc (Ag-Zn)	$\text{Zn} \rightarrow \text{ZnO}$	$\text{Ag}_2\text{O} \rightarrow \text{Ag}$	Zn^{2+}	Ag	KOH (aq)	Aerospace, submarines
Lithium Iron Phosphate (LiFePO4)	$\text{LiC}_6 \rightarrow \text{C} + \text{Li}^+$	$\text{LiFePO}_4 \rightarrow \text{FePO}_4$	Li^+	Li^0	Organic solvent with Li salt	EVs, solar storage

Lead Acid (Pb–acid)

Construction

A typical lead-acid battery consists of:

- **Negative plate (anode):** Spongy lead (Pb)
- **Positive plate (cathode):** Lead dioxide (PbO₂)
- **Electrolyte:** Dilute sulfuric acid (H₂SO₄, ~30–35%)
- **Separator:** Porous material to prevent direct contact between plates but allows ion flow
- **Container:** Usually made of hard rubber or plastic



Electrochemical Reactions

During DISCHARGE (The battery is powering a device)

In this mode, the battery acts as a **galvanic cell** (spontaneous reaction, producing electricity).

Anode (Oxidation, Negative Electrode):



Lead (Pb) is oxidized from 0 to +2.

Cathode (Reduction, Positive Electrode):



Lead in PbO₂ (oxidation state +4) is reduced to Pb²⁺ (in PbSO₄).

In discharge, both electrodes become coated with PbSO₄.

Overall Reaction During Discharge: $\text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$

Explanation:

- The **Pb at the anode loses electrons** → forms PbSO₄
- The **PbO₂ at the cathode gains electrons** → also forms PbSO₄
- Sulfuric acid concentration **decreases** as H₂SO₄ is consumed → battery voltage drops gradually

Result: Both electrodes are converted to **PbSO₄** and the electrolyte (H₂SO₄) gets **diluted** (less H₂SO₄, more water).

During CHARGING (The battery is consuming electricity)

In this mode, the battery acts as an **electrolytic cell** (non-spontaneous reaction, driven by an external power source).

This is the key part: The external charger *forces* electrons in the opposite direction. This reverses the chemical reactions and, crucially, **reverses the identity of the anode and cathode**.

Anode (Oxidation, Positive Electrode): This is now the electrode that was the cathode during discharge.



Lead in PbSO_4 (oxidation state +2) is oxidized to +4 (in PbO_2).

Cathode (Reduction, Negative Electrode): This is now the electrode that was the anode during discharge.



Lead in PbSO_4 (oxidation state +2) is reduced to 0 (in Pb metal).

Result: Electrodes return to their original forms: **Pb** (negative) and **PbO₂** (positive), and sulfuric acid concentration increases again.

Does the electrode material change?

Yes — both **chemically and physically**:

- **During discharge**, the electrodes are **converted to PbSO_4** .
- **During recharge**, they are **converted back to Pb and PbO_2** .

This transformation is **reversible**, which is what makes the lead-acid battery **rechargeable**.

However, over many cycles:

- PbSO_4 may not fully convert back.
- PbO_2 structure may degrade.
- Lead may fall off or crystallize.

These lead to **capacity fade** or **battery failure** over time.

Voltage

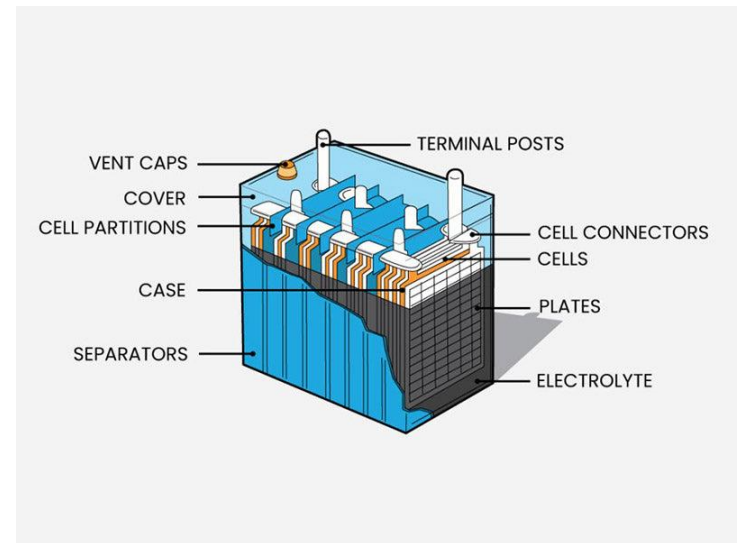
- **Single cell voltage:** $\sim 2\text{ V}$
- **Typical car battery:** 6 cells in series $\rightarrow 12\text{ V}$

Key Points

- Lead-acid batteries are **rechargeable** $\rightarrow$ secondary cells
- High **current output** $\rightarrow$ good for starting engines
- Heavy and low energy density $\rightarrow$ not ideal for portable electronics
- **Maintenance:** Water may need topping up in non-sealed batteries

Applications

- Automobiles (starter battery)
- UPS (Uninterruptible Power Supply)
- Emergency lights
- Solar energy storage





Lead Acid Batteries



Most important application of lead is in lead acid storage batteries for which more than 85% of the world production of lead is currently used.

Mode	Electrode Function	Electrode Name	Reaction (Lead-Acid)	Process
Discharge	Negative Terminal	Anode	$\text{Pb} \rightarrow \text{PbSO}_4$	Oxidation
	Positive Terminal	Cathode	$\text{PbO}_2 \rightarrow \text{PbSO}_4$	Reduction
Charge	Positive Terminal	Anode	$\text{PbSO}_4 \rightarrow \text{PbO}_2$	Oxidation
	Negative Terminal	Cathode	$\text{PbSO}_4 \rightarrow \text{Pb}$	Reduction

- The electrode materials themselves (Pb and PbO_2) are fixed.
- What changes is their state (oxidized or reduced) and their electrochemical role (anode or cathode).
- The negative terminal is Pb during discharge and becomes the cathode during charging, but it's still the same physical electrode — just undergoing a different reaction.

Detail description of Lead Acid (Pb–acid) battery for reading at home, not for exam purpose

Lead Acid Battery:

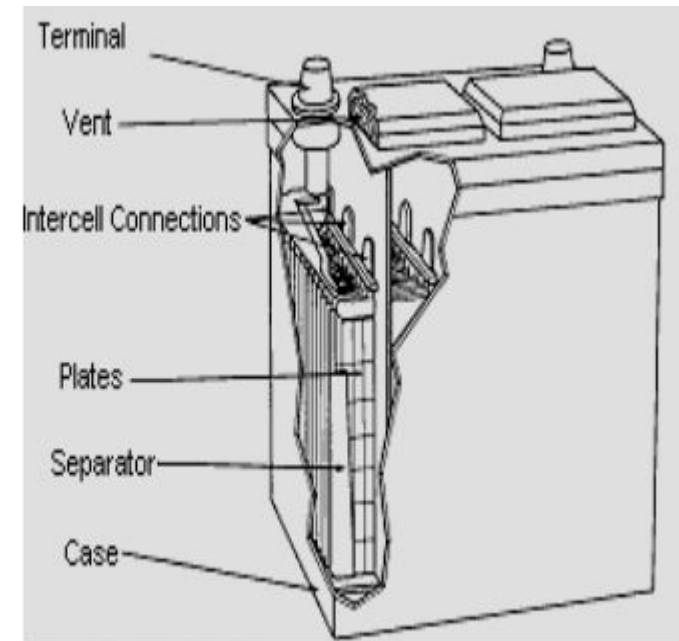
- Electrolyte for the most part distilled (pure) water, with some sulfuric acid mixed with the water.
- Electrodes must be of dissimilar metals.

Construction:

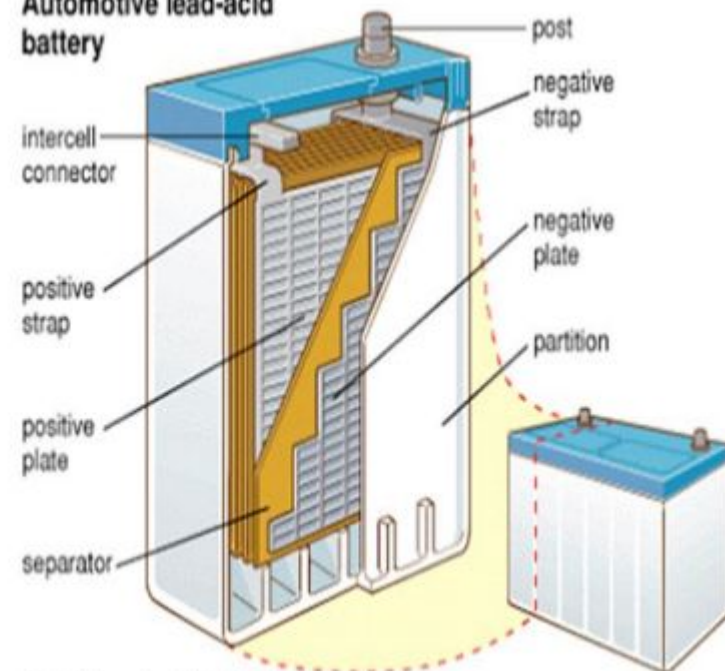
1.Separator: It is most important part of lead acid battery. Which separate the positive and negative plates from each other and prevents the short circuit? The separators must be porous so that the electrolyte may circulate between the plates . The separators must have higher insulating resistance and mechanical strength.

The material used for separators are wood, rubber, glass wood mate, pvc.

2. Electrolyte: in lead acid battery dilute sulphuric acid (H_2SO_4) is used as an electrolyte. For this purpose one part concentrated sulphuric acid is mixed with three parts of distilled water.



Automotive lead-acid battery



3. Container: Container is a box of vulcanized rubber, molded rubber, molded plastic, glass or ceramic , on the base of this box there are supports block on which the positive and negative plates are established. Thus between this supports there are grooves which works like a mud box. The active material separated from the plates get collected in this mud box and it cannot make the contact with the plates thus the internal faults due to the mud are avoided.

4. Cover of cell: In lead acid battery it is also made of the same material which is used is used for making container. It is used to cover the complete cell after the installation of the plates in it . it protects the cell from the dust as well as other external impurities.

5. Vent plug: The vent plug are provide in the cover plate of the cell which are used to fill up the electrolyte in the cell or the inspection of internal condition of the cell the vent plugs are also use for to exhaust the gases generated in the cell to the atmosphere.

6. Connecting bar: It works like a link and used to connect the two cells in series. Terminal of one cell and negative terminal of another cell.

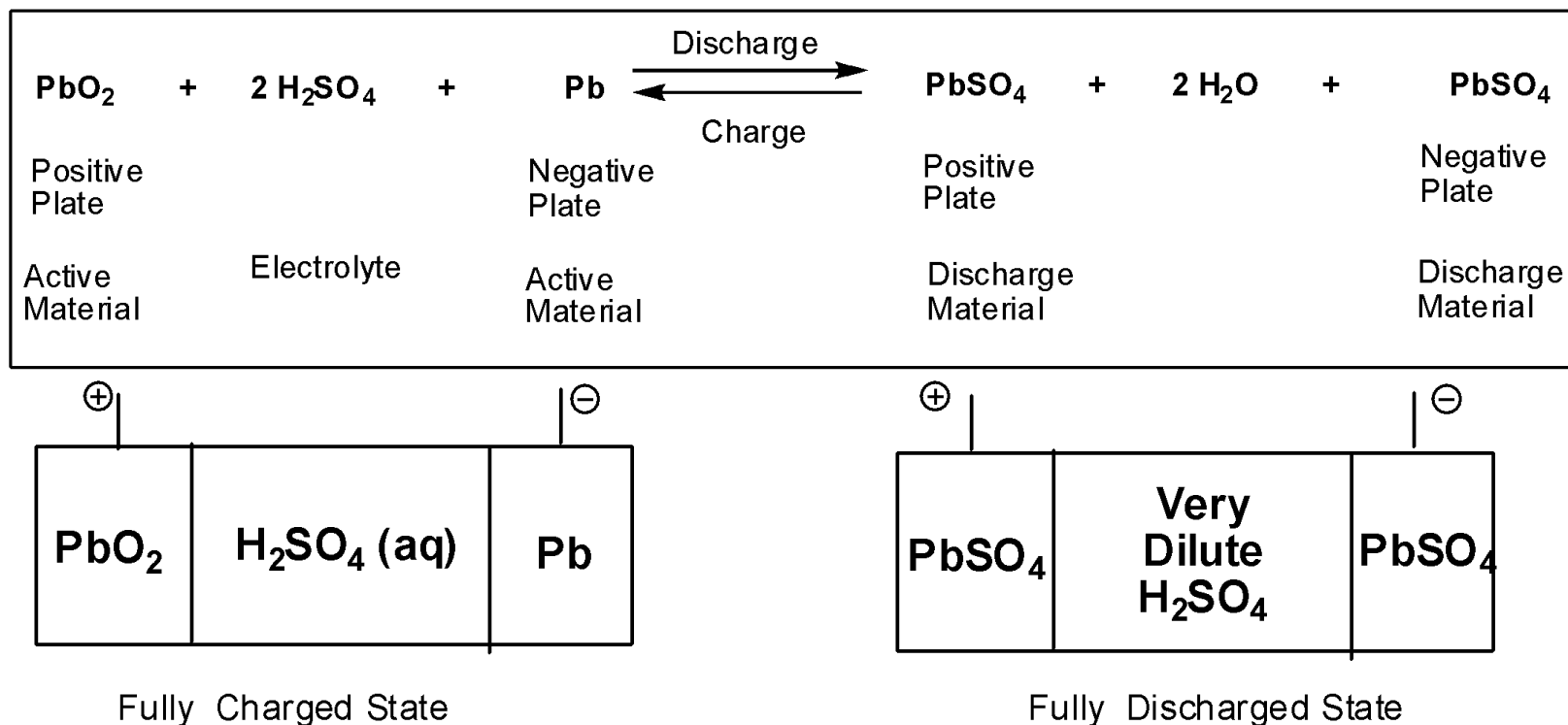
7 terminal posts: There are the terminals of the battery which are connected to charging circuit as well as the load. For identification the diameter of the positive terminal is design more as compared to the negative terminal.



Lead Acid Batteries



Most important application of lead is in lead acid storage batteries for which more than 85% of the world production of lead is currently used.



Structure of the Electrodes in a Lead-Acid Battery:

Negative Plate (Anode during discharge):

- **Active material:** **Sponge lead (Pb)** – porous to increase surface area.
- **Support structure:** **A lead alloy grid, often made of Pb mixed with small amounts of antimony (Sb) or calcium (Ca)** to improve:
 - Mechanical strength
 - Corrosion resistance
 - Conductivity

Positive Plate (Cathode during discharge):

- **Active material:** **Lead dioxide (PbO₂)** – a dark brown, brittle material.
- **Support structure:** **Also a lead alloy grid**, similar to the negative plate, but sometimes with different additives to withstand the more aggressive oxidative environment at the positive side.

How They're Assembled:

- The **active materials** (Pb and PbO₂) are **pasted onto or formed within** the grid.
- During **formation (initial charging of a new battery)**, the raw pasted material (often lead oxides or sulfates) is **electrochemically converted** into Pb on the negative plate and PbO₂ on the positive plate.

Why Not Use Pure Pb or PbO₂ Electrodes Alone?

Because:

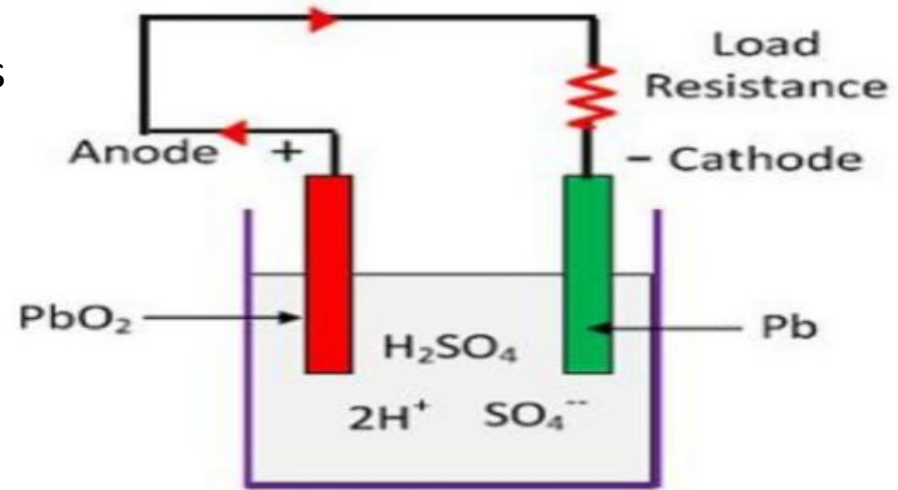
1. **Pure Pb is too soft** and lacks mechanical strength.
2. **PbO₂ is brittle** and would crack or fall apart under vibration or cycling.
3. The grid provides:
 - A **framework** to hold active material
 - A **low-resistance path** for electrons
 - Structural integrity under charge/discharge cycles

Working of lead acid battery:

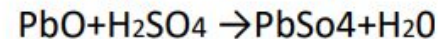
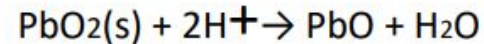
- Dilute sulfuric acid used for lead acid battery has ratio of acid: water = 1:3. This lead acid storage battery is formed by dipping lead peroxide plate and sponge lead plate in dilute sulfuric acid. A load is connected externally between these plates.

- During Discharging: In diluted sulfuric acid the molecules of the acid split into positive hydrogen ions and negative sulfate ions. The hydrogen ions when reach at PbO_2 plate, they receive electrons from it and become hydrogen atom which again attack PbO_2 and form PbO and H_2O (water). This PbO reacts with H_2SO_4 and forms PbSO_4 and H_2O .

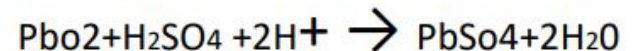
- SO_4^{--} ions are moving towards Pb plate where they give their extra electrons and react with Pb and forms PbSO_4



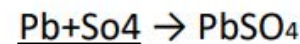
Positive plate reaction



The total reaction can be written as

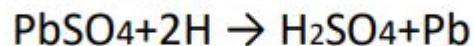


Negative plate reaction

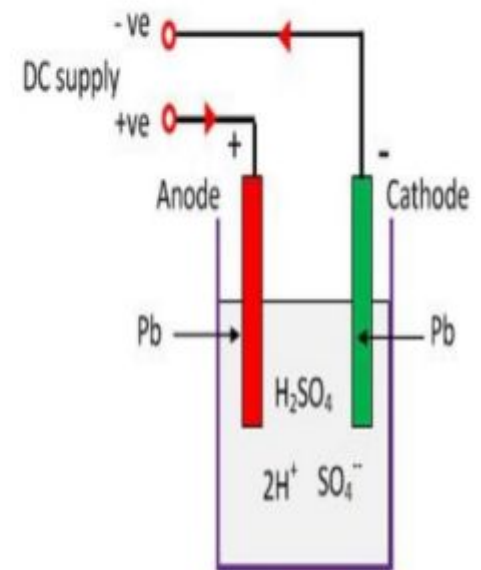


As H^+ ions take electron from PbO_2 plate and SO_4 ions give electrons to Pb plate, there would be an inequality of electrons between these two plate. Hence there would be flow of current through the external load between these plates for balancing this inequality of electron. This process is called discharging of lead acid battery.

During Charging: During discharging, the density of sulfuric acid falls but there still sulfuric acid exist in the solution. In this case Hydrogen ions being positive charged move to the cathode connected with $-ve$ terminal of DC source. Here each hydrogen ions take one electron from that and become hydrogen atom. These hydrogen atom then attack $PbSO_4$ and form lead and sulfuric acid.



Sulfate ions moves towards the anode connected with $+ve$ terminal of DC source where they will give up their extra electrons and become SO_4 and form lead peroxide and sulfuric acid.



Ventilation Requirements

- The oxygen and hydrogen gases released during the gassing phase of a typical flooded lead-acid battery recharge can be dangerous if allowed to **exceed 0.8 % (by volume) or 20 percent of the lower explosive range**. *Concentrations of hydrogen between 4 % and 74% are considered explosive (40,000 ppm and 740,000 ppm).*
- All lead acid power batteries give off gases when recharging and also for a period after the charge is completed.
 - A Concentration of hydrogen in excess of 4% (by volume). It is suggested that the concentration be controlled to a maximum of 2% (by volume).
- A typical lead acid motive power cell will, evolve approximately .016 cubic feet of hydrogen gas over A.H. overcharge.
- Since this gas is given off at the maximum rate at the end of the charging period, the following calculation assumes a charging current of 5% of the 6 hour A.H. capacity (C6) during this over charge period. (This charging current is excessive but has been used to take account of the worst case.)
- **Gas given off per hour per cell = $0.16 \times .05 = .0008$ C6 cu / ft. / cell / hr.**

Cell Construction

Ni–Cd battery

Anode (Negative electrode): Cadmium (Cd)

Cathode (Positive electrode): Nickel oxyhydroxide (NiO(OH))

Electrolyte: Aqueous potassium hydroxide (KOH) solution. The electrolyte (KOH) is not consumed in the *overall* reaction; it provides the OH⁻ ions that facilitate the reactions at each electrode.

Separator: Porous material (e.g., paper or synthetic) to prevent short circuit but allow ion flow

Container: Steel or plastic case

The overall reaction: $\text{Cd} + 2\text{NiOOH} + 2\text{H}_2\text{O} \rightleftharpoons \text{Cd}(\text{OH})_2 + 2\text{Ni}(\text{OH})_2$

During DISCHARGE (The battery is powering a device - Galvanic Cell): In this mode, the reaction is spontaneous.

Electrons flow from the negative terminal to the positive terminal through the external circuit.

• **Anode (Oxidation, Negative Electrode): $\text{Cd} + 2\text{OH}^- \rightarrow \text{Cd}(\text{OH})_2 + 2\text{e}^-$**

• **What's happening:** Cadmium metal (Cd, oxidation state 0) is oxidized to Cadmium ion (oxidation state +2) in Cadmium Hydroxide.

• This is the source of electrons.

• **Cathode (Reduction, Positive Electrode): $2\text{NiOOH} + 2\text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{Ni}(\text{OH})_2 + 2\text{OH}^-$**

• **What's happening:** Nickel in NiOOH (oxidation state +3) is reduced to Nickel in Ni(OH)₂ (oxidation state +2).

• This is where electrons are consumed.

2. During CHARGING (The battery is consuming electricity - Electrolytic Cell)

This is the key part, exactly as with the lead-acid battery. The external charger *forces* electrons in the opposite direction, reversing the chemical reactions and **swapping the anode/cathode labels**.

Anode (Oxidation, Positive Electrode): This is now the electrode that was the cathode during discharge.



What's happening: Nickel in $\text{Ni}(\text{OH})_2$ (oxidation state +2) is *oxidized* to +3 (in NiOOH).

This is where electrons are *forced out* of the battery to the positive terminal.

Cathode (Reduction, Negative Electrode): This is now the electrode that was the anode during discharge.



What's happening: Cadmium in $\text{Cd}(\text{OH})_2$ (oxidation state +2) is *reduced* to Cadmium metal (oxidation state 0). This is where electrons are *forced into* the battery from the negative terminal.

Explanation:

- External current forces **$\text{Cd}(\text{OH})_2$ back to Cd** and **$\text{Ni}(\text{OH})_2$ back to $\text{NiO}(\text{OH})$** .
- Battery **regains its capacity** to supply current.

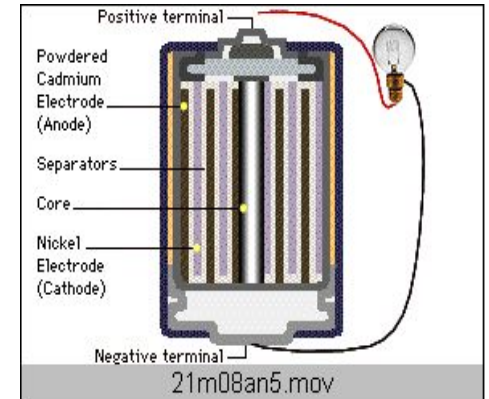
Mode	Electrode Function	Electrode Name	Reaction	Process
Discharge	Negative Terminal	Anode	$\text{Cd} \rightarrow \text{Cd}(\text{OH})_2$	Oxidation
		Cathode	$\text{NiOOH} \rightarrow \text{Ni}(\text{OH})_2$	Reduction
Charge	Positive Terminal	Anode	$\text{Ni}(\text{OH})_2 \rightarrow \text{NiOOH}$	Oxidation
		Cathode	$\text{Cd}(\text{OH})_2 \rightarrow \text{Cd}$	Reduction

Voltage

- **Single cell voltage:** $\sim 1.2\text{ V}$
- Multiple cells can be connected in series to get higher voltages.

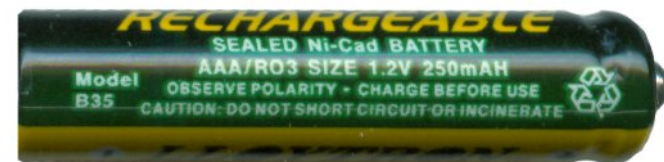
4. Key Points

- Ni-Cd is **rechargeable** $\rightarrow$ secondary cell
- High **discharge rate** and good **low-temperature performance**
- **Memory effect:** Can lose capacity if repeatedly partially discharged and recharged
- Electrolyte (KOH) is **not consumed**, only facilitates ion transport



5. Applications

- Power tools (drills, saws)
- Emergency lights and flashlights
- Cordless phones
- Toys and portable electronic devices
- Aerospace and military equipment



What is Memory Effect?

- The **memory effect** (or “voltage depression”) happens mainly in **Nickel–Cadmium (Ni–Cd)** batteries.
- If a Ni–Cd cell is **repeatedly partially discharged and recharged**, it can “remember” the shorter cycle and lose usable capacity.
- Example:

If you always discharge only 30% before recharging → over time, the battery “thinks” its full capacity is just 30%, so voltage drops earlier.

Why does it happen in Ni–Cd?

- In Ni–Cd, **cadmium at the anode** (negative electrode) participates in the redox reaction:
$$\text{Cd} + 2\text{OH}^- \leftrightarrow \text{Cd}(\text{OH})_2 + 2\text{e}^-$$
- When the battery is **charged and discharged fully**, cadmium tends to deposit back as **fine crystals** → which is good for performance.
- But when the cell is **only shallow-cycled** (not fully discharged before charging again) → the deposition is **uneven**, and **larger Cd crystals** start forming.
- These **large crystals** have less active surface area → higher internal resistance, lower usable capacity.
- This is what we call the **memory effect**
- The result: apparent loss of capacity (though not truly gone, but “blocked” by crystal formation).

Lithium-ion battery



Lithium-ion battery

Fundamental Concept Recap: The rule remains: The anode is where oxidation occurs, and the cathode is where reduction occurs. Their identities swap during charge and discharge.

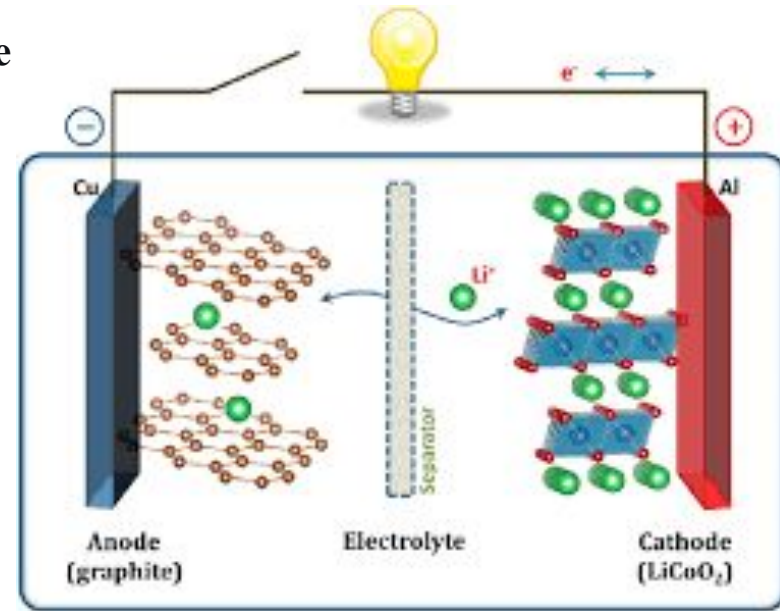
Fundamental Concept Recap: The rule remains: The anode is where oxidation occurs, and the cathode is where reduction occurs. Their identities swap during charge and discharge.

However, with Li-ion batteries, the terminology is often fixed by the manufacturer:

- **Cathode:** The lithium metal oxide (e.g., LiCoO_2). This is the "positive" electrode.
- **Anode:** The carbon-based material (e.g., Graphite). This is the "negative" electrode.

The Electrochemical Reactions (Example: LiCoO_2 / Graphite System)

- **Positive Electrode (Physical Component):** Lithium Cobalt Oxide (LiCoO_2)
- **Negative Electrode (Physical Component):** Graphite (C_6)
- **Electrolyte:** Lithium salt (e.g., LiPF_6) in an organic solvent.
- **Mechanism:** Lithium ions (Li^+) shuttle back and forth between the electrodes, intercalating (embedding) into their crystalline structures. This is why it's called a "rocking chair" battery.



1. **During DISCHARGE (Powering a device - Galvanic Cell):** This is a spontaneous process. The negative electrode has a high chemical potential for Li, and the positive electrode has a low one. Li^+ moves through the electrolyte, and electrons power your device through the external circuit.

• **Anode (Oxidation, Negative Electrode - Graphite):** $\text{LiC}_6 \rightarrow \text{C}_6 + \text{Li}^+ + \text{e}^-$

What's happening: Lithium atoms intercalated between graphene layers are oxidized into Li^+ ions and electrons. The Li^+ ions travel into the electrolyte, and the electrons travel through the external circuit to the positive side.

• **Cathode (Reduction, Positive Electrode - LiCoO_2):**

$\text{Li}_{1-x}\text{CoO}_2 + x\text{Li}^+ + x\text{e}^- \rightarrow \text{LiCoO}_2$ (or more precisely for the discharge step: $\text{Li}_{0.5}\text{CoO}_2 + 0.5\text{Li}^+ + 0.5\text{e}^- \rightarrow \text{LiCoO}_2$)

What's happening: The Li^+ ions from the electrolyte and the electrons from the external circuit are absorbed, reducing the Cobalt and intercalating Li^+ into the metal oxide structure. **Overall Discharge Reaction:** $\text{LiC}_6 + \text{Li}_{0.5}\text{CoO}_2 \rightarrow \text{C}_6 + \text{LiCoO}_2$

2. **During CHARGING (Consuming electricity - Electrolytic Cell):** The external charger

applies a voltage, *forcing* the reaction to run in reverse. This reverses the current direction

and **swaps the electrochemical functions** of the electrodes.

• **Anode (Oxidation, Positive Electrode - LiCoO_2):**

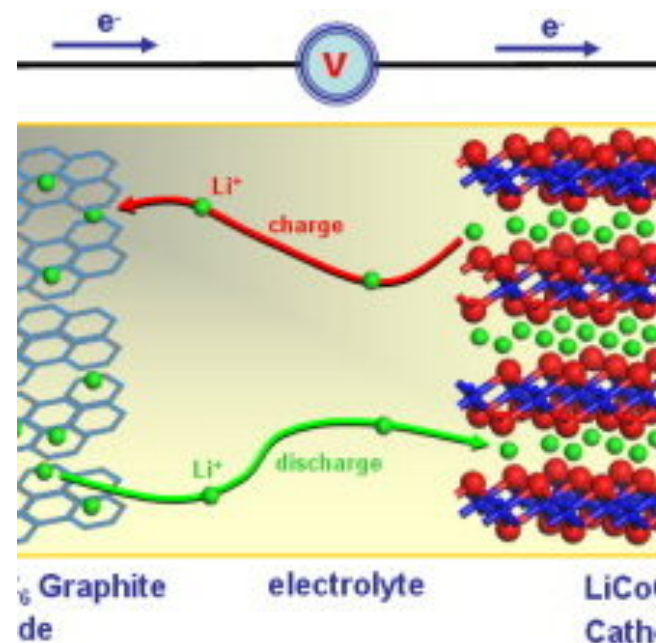
• $\text{LiCoO}_2 \rightarrow \text{Li}_{1-x}\text{CoO}_2 + x\text{Li}^+ + x\text{e}^-$ (e.g., $\text{LiCoO}_2 \rightarrow \text{Li}_{0.5}\text{CoO}_2 + 0.5\text{Li}^+ + 0.5\text{e}^-$)

What's happening: The charger "pulls" electrons from the positive electrode, oxidizing it and kicking Li^+ ions out of the metal oxide structure and into the electrolyte.

• **Cathode (Reduction, Negative Electrode - Graphite):** $\text{C}_6 + \text{Li}^+ + \text{e}^- \rightarrow \text{LiC}_6$

• **What's happening:** The charger "pushes" electrons to the negative electrode. Li^+ ions from the electrolyte are simultaneously attracted there, where they are reduced and intercalate between the graphene layers.

Overall Charge Reaction: $\text{C}_6 + \text{LiCoO}_2 \rightarrow \text{LiC}_6 + \text{Li}_{0.5}\text{CoO}_2$



The Anode and Copper (Cu) Current Collector: The graphite-based anode is deposited on a Copper (Cu) foil.

Reason: Copper is an excellent conductor and is **electrochemically stable** in the potential range where the graphite anode operates (approximately 0.1 to 0.2 V vs. Li/Li⁺). At these low potentials, aluminum would form a lithium-aluminum alloy and corrode, which would be disastrous for the battery's performance and safety.

The Cathode and Aluminum (Al) Current Collector: The LiCoO₂-based cathode is deposited on an Aluminum (Al) foil.

Reason: Aluminum is an excellent conductor and forms a protective, stable passivation layer of aluminum oxide (Al₂O₃) on its surface. This layer prevents further reaction and corrosion at the high operating potentials of the cathode (around 3.9 V vs. Li/Li⁺). At these high voltages, copper would readily oxidize and dissolve, degrading the battery

Component	Active Material	Current Collector	Why this Collector?
Anode	Graphite (LiC ₆)	Copper (Cu)	Stable at low potentials; does not alloy with lithium.
Cathode	Lithium Cobalt Oxide (LiCoO ₂)	Aluminum (Al)	Forms a passivating oxide layer; stable at high potentials.

Key Features

- **Voltage per cell:** $\sim 3.6\text{--}3.7\text{ V}$
- **High energy density** $\rightarrow$ lightweight, ideal for portable electronics
- **No memory effect** (unlike Ni–Cd)
- Requires **protection circuit** for overcharge, over-discharge, and short-circuit prevention
- Lithium ions **intercalate into graphite** rather than forming metallic lithium $\rightarrow$ improves safety

Why Li-ion has *No Memory Effect*

- In **Li-ion**, energy storage works by **intercalation of Li^+ ions** into the graphite anode and lithium metal oxide cathode.
- This is a **reversible solid-state process** — no large crystal growth occurs as in Ni–Cd.
- Therefore, Li-ion cells don't “remember” partial discharge cycles.
- You can **recharge them at any point** (10%, 50%, 80%) without reducing long-term capacity.

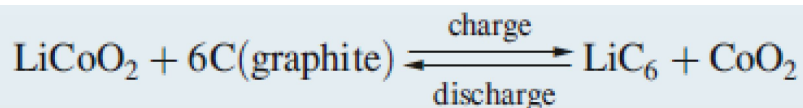
4. Applications

- Mobile phones, laptops, tablets
- Electric vehicles (EVs)
- Power tools
- Drones and wearable electronics
- Energy storage systems



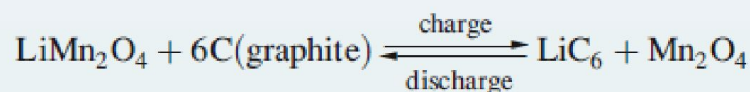
Lithium-ion battery

An important advance in battery technology has been the development of rechargeable, high energy-density lithium-ion batteries, first introduced to the commercial market in 1991. Ten billion US dollars worth of lithium-ion batteries were sold in 2008, and the market continues to grow. The lithium-ion battery has a cell potential of 3.6 V and consists of a positive LiCoO_2 electrode separated from a graphite electrode by a solid electrolyte across which Li^+ ions can migrate when the cell is charging. In commercial lithium-ion batteries, the electrolyte is usually LiPF_6 in an alkyl carbonate material. Lithium-ion batteries are manufactured in a discharged state. Solid LiCoO_2 adopts an $\alpha\text{-NaFeO}_2$ structure type in which the O atoms are approximately cubic close-packed. The octahedral holes are occupied by M(I) or M'(III) (Li^+ or Co^{3+} in LiCoO_2) in such a way that the different metal ions are arranged in layers. During charging, Li^+ ions move out of these layers, are transported across the electrolyte, and are intercalated by the graphite (see Section 14.4). During discharge of the cell, the Li^+ ions return to the metal oxide lattice. The cell reaction can be represented as follows:



The cobalt centres are redox active, being oxidized from Co(III) to Co(IV) as Li^+ is removed from LiCoO_2 . The crucial factor in lithium-ion batteries is that both electrodes are able to act as hosts for Li^+ ions. Rechargeable, lithium-ion batteries now dominate the market for small electronic devices such as laptop computers, mobile phones, iPods and MP3 players, and in electric bicycles. In 2005, Sony introduced a new generation of lithium-ion batteries (the Nexelion battery) in which the mixed metal oxide $\text{Li}(\text{Ni},\text{Mn},\text{Co})\text{O}_2$ replaces the all-cobalt LiCoO_2 electrode, and a tin-based electrode replaces graphite.

A disadvantage of lithium-ion batteries containing cobalt is their relatively high cost. Current research strategies are aimed at finding replacement electrode materials both to increase battery performance and to reduce cost. Two contenders are LiMn_2O_4 and LiFePO_4 . LiMn_2O_4 has a spinel structure (see Box 13.7) and when coupled with a graphite electrode forms a lithium-ion battery, the cell reaction of which is summarized below:

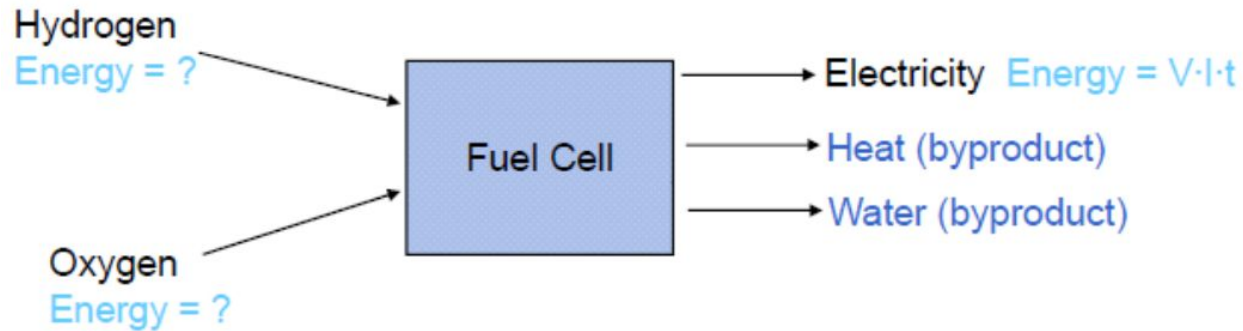


Which one is the best rechargeable battery among all

- **For portable electronics: Li-ion** → high energy density, lightweight.
- **For long life and safety: LiFePO_4** → lower energy density but much safer and longer cycle life.
- **For automotive starter / heavy power: Lead-acid** → cheap and high current output.
- **For industrial/military: Ni-Cd or Ag-Zn** → high reliability, can handle extreme conditions.

Fuel Cells

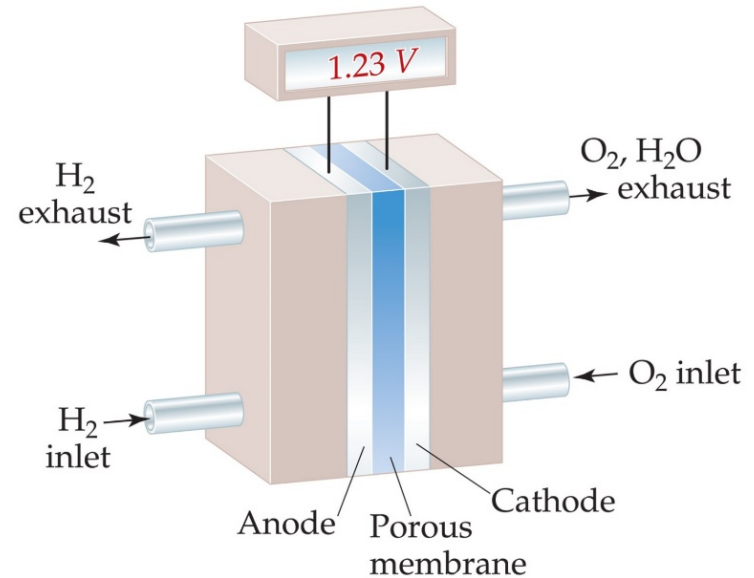
**Basic energy conversion of a fuel cell can be described as:
Chemical energy of fuel = Electrical energy + Heat energy**



Fuel Cells

What is a Fuel Cell?

- A fuel cell is an electrochemical device that converts the chemical energy of a fuel (like hydrogen) and an oxidant (like oxygen) directly into electricity, heat, and water. Unlike batteries, which store energy, fuel cells generate electricity continuously as long as fuel and oxidant are supplied.
- When a fuel is burned, the energy created can be converted to electrical energy.
- Usually, this conversion is only 40% efficient, with the remainder lost as heat.
- Fuel cells are NOT batteries; the source of energy must be continuously provided.



- ✓ A *fuel cell* converts a chemical fuel, such as hydrogen (used for larger power requirements) or methanol (a convenient fuel for small applications), directly into electrical power, using O₂ or air as the oxidant. As power sources, fuel cells offer several advantages over rechargeable batteries or combustion engines, and their use is steadily increasing.

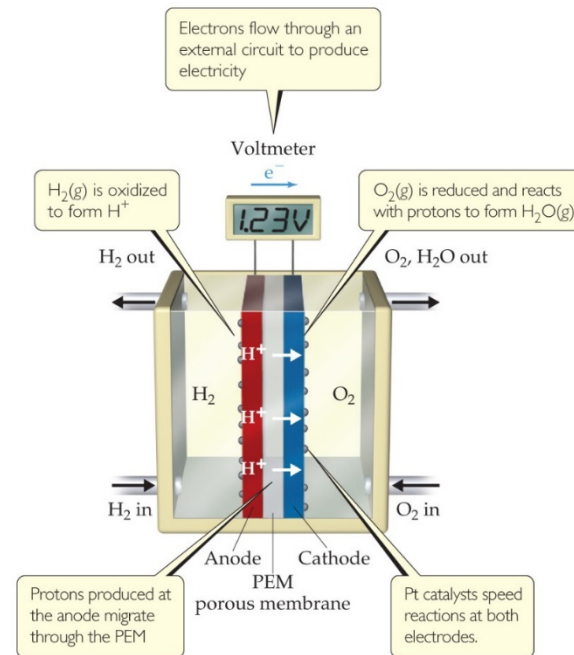
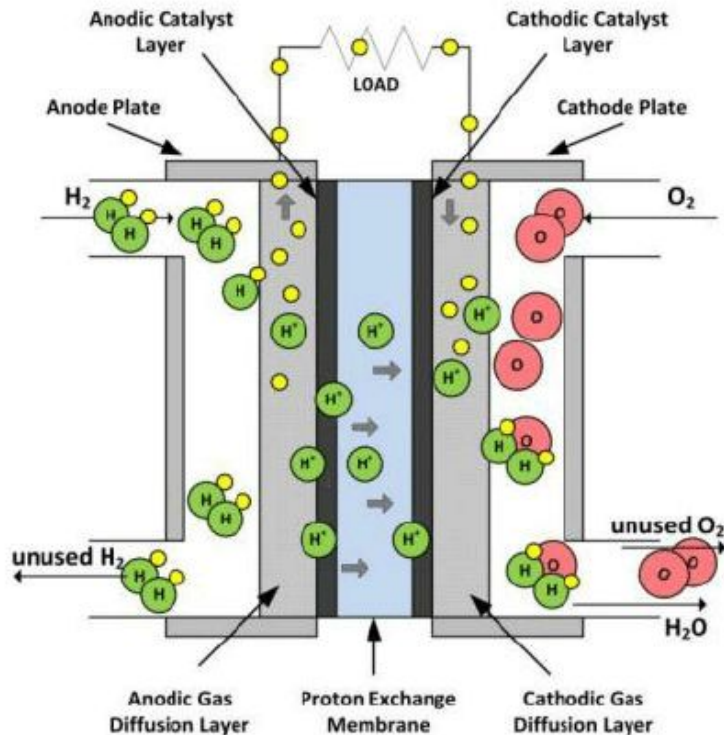
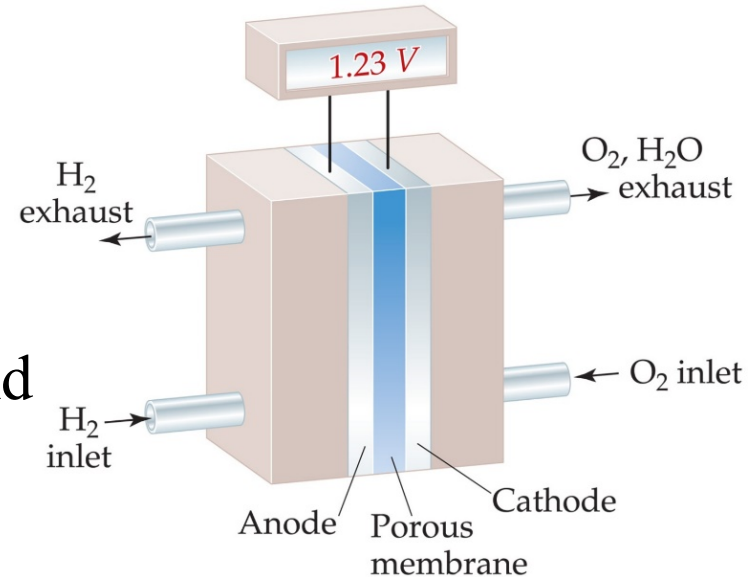
First developed by William Grove In 1839, Grove was experimenting on electrolysis (the process by which water is split into hydrogen and oxygen by an electric current), when he observed that combining the same elements could also produce an electric current

Fuel Cell types

Fuel Cell Type	Anode Reaction	Cathode Reaction	Electrolyte	Anode–Cathode Separation	Applications
PEMFC (Proton Exchange Membrane)	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\frac{1}{2} \text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$	Polymer proton-conducting membrane (Nafion)	Polymer membrane $\rightarrow$ allows only H^+ to pass, keeps H_2 and O_2 separate	Cars (Toyota Mirai), portable power
AFC (Alkaline Fuel Cell)	$\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$	$\frac{1}{2} \text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^-$	Aqueous KOH (alkaline)	Porous separator soaked in KOH $\rightarrow$ keeps H_2 and O_2 apart	Spacecraft (Apollo), military backup
PAFC (Phosphoric Acid Fuel Cell)	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$\frac{1}{2} \text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$	Liquid phosphoric acid (H_3PO_4)	Porous matrix holds H_3PO_4 and separates gases	Stationary power, hospitals, data centers
MCFC (Molten Carbonate Fuel Cell)	$\text{H}_2 + \text{CO}_3^{2-} \rightarrow \text{H}_2\text{O} + \text{CO}_2 + 2\text{e}^-$	$\frac{1}{2} \text{O}_2 + \text{CO}_2 + 2\text{e}^- \rightarrow \text{CO}_3^{2-}$	Molten carbonate salts ($\text{Li}_2\text{CO}_3/\text{K}_2\text{CO}_3$)	Porous ceramic matrix retains molten salts, prevents gas mixing	Power plants, industrial CHP
SOFC (Solid Oxide Fuel Cell)	$\text{H}_2 + \text{O}^{2-} \rightarrow \text{H}_2\text{O} + 2\text{e}^-$	$\frac{1}{2} \text{O}_2 + 2\text{e}^- \rightarrow \text{O}^{2-}$	Solid ceramic oxide (YSZ)	Dense ceramic electrolyte conducts O^{2-} only, gases in separate chambers	Power generation, APUs
DMFC (Direct Methanol Fuel Cell)	$\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 6\text{H}^+ + 6\text{e}^-$	$1.5 \text{O}_2 + 6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\text{O}$	Polymer proton-conducting membrane	Polymer membrane (like PEMFC), but methanol crossover is an issue	Portable electronics, small systems
RFC (Regenerative Fuel Cell)	$\text{H}_2\text{O} \rightleftharpoons \text{H}_2 + \frac{1}{2} \text{O}_2$ (reversible)	Same as PEMFC	Same as PEMFC	PEM or solid electrolyte separates gases, can switch modes	Space missions, renewable energy storage

Hydrogen Fuel Cells

- In this cell, hydrogen and oxygen form water.
- The cells are twice as efficient as combustion.
- The cells use hydrogen gas as the fuel and oxygen from the air.



Fuel Cells

Basic Operating Principle

The core principle is the reverse of electrolysis. A fuel cell uses hydrogen and oxygen to create electricity through an electrochemical reaction, without combustion.

•**Overall Reaction:** $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{Electricity} + \text{Heat}$

•**Core Components:**

- Anode:** The negative post where hydrogen gas is supplied.
- Cathode:** The positive post where oxygen (from air) is supplied.
- Electrolyte:** A specialized material (e.g., a polymer membrane) that conducts positively charged ions (protons) and blocks electrons.
- Catalyst:** Usually Platinum, which speeds up the reactions at the electrodes.

Step-by-Step Process (in a PEM Fuel Cell)

(PEM stands for Proton Exchange Membrane fuel cell)

The most common type for vehicles and portable applications is the **Proton Exchange Membrane (PEM) Fuel Cell**.

- 1. Hydrogen Supply:** Hydrogen gas (H_2) is fed into the anode.
- 2. Splitting Hydrogen:** At the anode, the hydrogen molecules come into contact with the catalyst and split into protons (H^+) and electrons (e^-). $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
- 3. Ion Movement:** The protons pass through the electrolyte membrane to the cathode. The membrane blocks the electrons.
- 4. Electron Flow:** The electrons cannot pass through the membrane and are forced to take an external path, creating a flow of direct electric current. This current can power a motor, device, or be stored in a battery.
- 5. Oxygen Supply:** Oxygen from the air (O_2) is fed into the cathode.
- 6. Water Formation:** At the cathode, the protons, electrons (which have done their work), and oxygen combine to form pure water (H_2O). $4\text{H}^+ + 4\text{e}^- + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- 7. Heat Release:** The reaction also releases heat, which must be managed.

Overall Reaction: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{Electricity} + \text{Heat}$

Electrode	Material (Typical)	Function
Anode	Pt/C (Platinum on Carbon)	Oxidation of H ₂
Cathode	Pt/C (more Pt than anode usually)	Reduction of O ₂

1. At the Anode (H₂ side):

- The reaction is:

$$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^- \text{ (oxidation)}$$
- This requires breaking the **H–H bond** and facilitating **proton release**.
- **Why Pt?**
 - Pt strongly **adsorbs H₂**, but not too strongly (ideal binding).
 - It allows **efficient splitting of H₂** into protons and electrons.
 - Has **fast hydrogen oxidation reaction (HOR)** kinetics.

2. At the Cathode (O₂ side):

- The reaction is:

$$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O} \text{ (reduction)}$$
- This is a **much slower reaction** than the anode.
- **Why Pt?**
 - Pt is one of the few catalysts that can **efficiently reduce O₂** at reasonable temperatures.
 - Oxygen reduction reaction (ORR) is complex (multi-step, multi-electron).
 - Pt catalyzes this reaction with relatively good efficiency (though still sluggish compared to HOR).

Why does H₂ split?

At the anode, a catalyst (usually platinum) facilitates the **oxidation of hydrogen**:

1. The hydrogen molecule adsorbs onto the catalyst surface.
2. The H–H bond weakens.
3. Each H atom loses an electron → forming H⁺ ions.

This happens because the catalyst **lowers the activation energy** for the reaction, making the electron transfer feasible.

Driving Force

The driving force is **electrochemical potential difference**, which is a combination of:

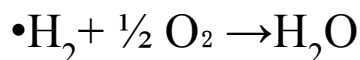
1. Chemical potential difference:

1. Hydrogen gas is at high chemical potential at the anode.
2. Oxygen (or oxidizer) at the cathode has a strong tendency to gain electrons and form water.

2. Electrical potential difference:

1. The external circuit allows electrons to flow from high potential (anode) to low potential (cathode).
2. This flow is spontaneous because the **overall Gibbs free energy change (ΔG) for the cell reaction is negative**.

The net cell reaction is:



• $\Delta G^\circ < 0 \rightarrow$ the reaction is spontaneous.

• This spontaneity is what “pulls” H₂ to split into H⁺ and e[−].

How much voltage does the H₂ Fuel cell generate practically

The voltage required to *operate* a hydrogen fuel cell is different from the voltage it *produces*.

A single Hydrogen Fuel Cell requires a voltage of **at least 0.6 to 0.7 Volts** to overcome its internal losses and start operating effectively. The theoretical maximum voltage a single H₂/O₂ fuel cell can produce is **~1.23 Volts** under ideal, reversible conditions at 25°C. However, in reality, this is never achieved due to three types of irreversible losses:

1.Activation Losses (Voltage Drop at Low Current):

- 1. What it is:** The energy required to start the electrochemical reactions at the electrodes (especially the oxygen reduction at the cathode).
- 2. Effect:** Causes an immediate drop from 1.23V to around **0.9 - 1.0V** as soon as any current is drawn.

2.Ohmic Losses (Voltage Drop Proportional to Current):

- 1. What it is:** Resistance to the flow of ions through the electrolyte and electrons through the various components.
- 2. Effect:** Causes a linear voltage drop as more current is drawn. This is the primary region of efficient operation.

3.Concentration Losses (Voltage Drop at High Current):

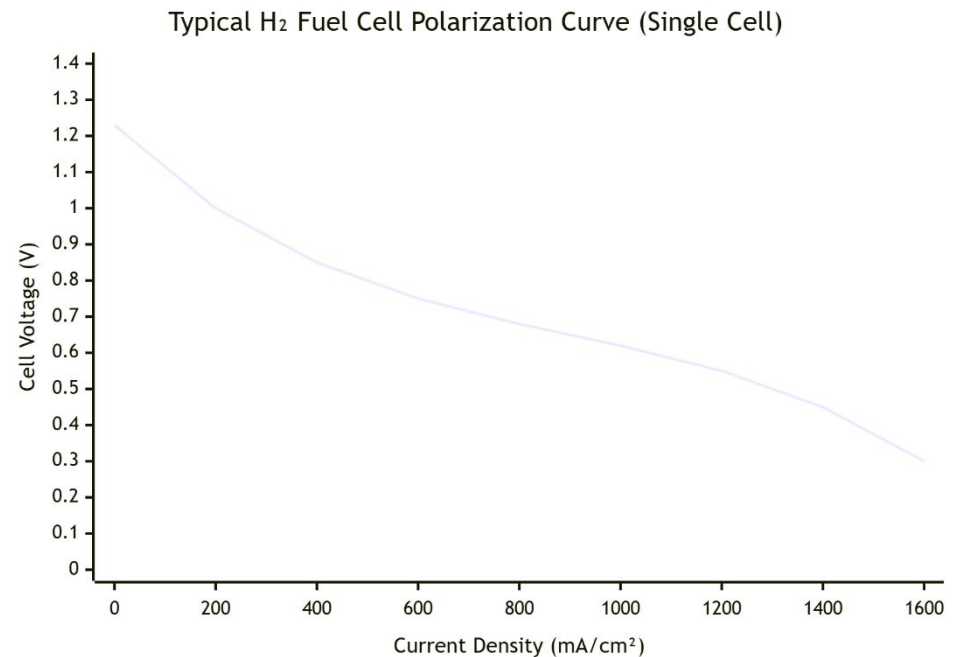
- 1. What it is:** At high currents, reactants can't diffuse to the reaction sites fast enough, and products (water) can't be removed quickly, starving the cell.
- 2. Effect:** Causes a rapid voltage drop, leading to a collapse in power.

1. As soon as you start an engine. You Close the Circuit (e.g., start the motor): A demand for current is created.

2. Initial Voltage Drop: As you correctly saw from the graph, the voltage *instantaneously* drops from the open-circuit voltage ($\sim 1.0\text{V}$) due to activation and ohmic losses. Let's say it settles at its operating point of **0.7 V**.

3. The Crucial "Refueling" Process (This is what you're missing):

1. This voltage drop and current flow is a *signal* to the electrochemical system.
2. To sustain this current flow, hydrogen and oxygen are continuously **consumed at the electrodes**.
3. The fuel cell system's balance-of-plant (pumps, injectors) responds by **supplying fresh hydrogen and air at a rate that matches the electrical demand**.
4. As long as fuel and oxidant are supplied, the electrochemical reactions continue at a steady rate, **maintaining a stable electron flow and a stable operating voltage** (e.g., 0.7V)



What Happens if You Draw *Too Much Current*?

This is where your logic becomes partially correct in a failure scenario. If you try to draw an excessively high current (e.g., by short-circuiting the stack or demanding extreme power):

1. You move far to the right on the polarization curve into the "concentration loss" region. The reactants cannot be supplied to the reaction sites fast enough. The voltage **plummets dramatically** (as the graph shows). If the voltage drops too low (e.g., below 0.4-0.5 V), it can cause permanent damage. This is known as **fuel cell flooding** (water isn't removed) or **starving** (not enough fuel).
2. In a real system, the **voltage monitoring system would detect this** and either open the circuit or limit the current to protect the stack, preventing it from reaching zero.

Voltage in Practical Applications: The Fuel Cell Stack

Since a single cell produces less than 1 Volt, they are combined in series to form a **Fuel Cell Stack** to achieve a useful voltage.

- **Cars & Buses:** A typical automotive fuel cell stack has **300 to 400 individual cells**.
 - If each cell operates at 0.7V, the total stack voltage is $300 \times 0.7V = \mathbf{210\ VDC}$.
 - This high DC voltage is then used to power an electric motor, often through a DC-DC converter and an inverter.
- **Stationary Power:** Large stationary stacks for backup or primary power are also designed to output standard voltages like **48 VDC, 120 VDC, or 240 VDC**.

Key Characteristics

- **Energy Efficiency:** Very high. 40-60% efficient in converting fuel to electricity (compared to ~20-30% for a typical internal combustion engine). When utilizing the waste heat (cogeneration), efficiency can exceed 85%.
- **Power Density:** High, making them suitable for vehicles where space and weight are critical.
- **Response Time:** Very fast power response (seconds), ideal for dynamic applications like cars.
- **Operating Temperature:** Varies by type. PEM cells run at low temps (50-100°C), while Solid Oxide cells run very hot (500-1000°C).
- **Lifespan:** Currently, fuel cell stacks are designed for lifetimes comparable to vehicle engines (e.g., 5,000-8,000 operating hours)

Advantages:

- **High Efficiency:** Fuel cells are not limited by the Carnot cycle (a limitation for heat engines). They can achieve 40-60% energy conversion efficiency, and over 80% if waste heat is captured (cogeneration).
- **Clean Energy:** With pure hydrogen, the only emission is water vapor. This significantly reduces greenhouse gases and air pollutants.
- **Reliability & Quiet Operation:** Few moving parts lead to quiet, reliable, and low-maintenance operation.
- **Fuel Flexibility:** While hydrogen is ideal, some fuel cells (SOFC, MCFC) can use natural gas, biogas, or methanol.
- **Scalability:** Fuel cells can be designed for a wide range of sizes, from powering a laptop to a small town.

Disadvantages:

- **High Cost:** The use of precious metals (like platinum) as catalysts and specialized materials makes fuel cells expensive.
- **Hydrogen Storage & Production:** Hydrogen is difficult to store and transport efficiently (it requires high pressure or very low temperatures). Most hydrogen today is produced from natural gas ("grey hydrogen"), which is not carbon-free.
- **Durability and Lifespan:** Performance can degrade over time, especially with impurities in the fuel or under varying load conditions.
- **Lack of Infrastructure:** Hydrogen refueling stations are not yet widely available, hindering the adoption of fuel cell vehicles.

Applications

Fuel cells are versatile and are being deployed in several key sectors:

- **Transportation:**
 - **Cars and Buses:** Fuel Cell Electric Vehicles (FCEVs) offer long range and fast refueling times (3-5 minutes) compared to battery-electric vehicles.
 - **Trains and Trucks:** For heavy-duty transport where battery weight becomes prohibitive.
 - **Drones and Maritime Vessels:** For longer flight/operation times.
- **Stationary Power:**
 - **Backup Power:** For critical facilities like data centers, hospitals, and telecommunications towers.
 - **Primary Power:** For remote or off-grid locations (e.g., cabins, weather stations).
 - **Combined Heat and Power (CHP):** Systems for large buildings that generate both electricity and useful heat.
- **Portable Power:**
 - **Military:** To power soldier-borne equipment, remote sensors.
 - **Consumer Electronics:** Potentially for laptops and mobile phones, though this market is still nascent.